

Session C1 — Electronegativity, Bond Character & the Three Bond Types

Duration: 45 minutes (strict) — **this is an ambitious scope for 45 minutes; see the pacing note below for what to cut first if you're running behind.** **Replaces:** the earlier "Periodic Trends Recap" version of today's session — this is the rebuilt version covering the fuller topic list. **Grade-9 foundation used:** Ch9 *Atomic Foundations of Matter* (ionic bonding, ion tables, criss-cross method, covalent naming with prefixes — section 9.4.2) **Grade-11 depth:** NCERT Class 11 Chemistry Ch4 *Chemical Bonding and Molecular Structure*, section 4.3.6 "Polarity of Bonds"; NCERT Class 11 Ch3 (electronegativity values); NCERT Class 10 Ch3 *Metals and Non-metals* (properties of pure metals, alloys) **Interactive tool (4 modules now): "Trends to Bonds"** — <https://claude.ai/artifact/JLvqSjiFUK7GfAfcCQVhxg>

- Module 1: Periodic Trends Explorer
- Module 2: Build an Ionic Bond (electron transfer + criss-cross, animated)
- Module 3 (**new**): Bond Character spectrum — pick any two atoms, see ΔEN placed on a nonpolar→polar→ionic spectrum
- Module 4 (**new**): Naming Covalent Compounds — 10-question prefix-naming practice, both directions

Lesson Plan

Objectives — by the end of this session the student should be able to:

1. State what electronegativity difference (ΔEN) predicts, and place a given bond on the nonpolar-covalent → polar-covalent → ionic spectrum.
2. Describe metallic bonding (electron-sea model) and list the properties of pure metals it explains.
3. Write ionic formulas — including with polyatomic ions — using the criss-cross method.
4. Name binary covalent compounds using Greek prefixes, including the three "gotcha" rules (dropped vowel, hydrogen-first, common names).

Time	Segment
0–3 min	Warm-up / hook
3–10 min	The framework: Electronegativity & Bond Character
10–15 min	Metallic bonds, properties of pure metals, alloys (brief)
15–24 min	Ionic bonds, ionic ratios, polyatomic ions, ionic + polyatomic combos
24–34 min	Covalent bonds + naming with prefixes
34–42 min	Interactive: Bond Character + Naming Covalent modules
42–45 min	Wrap-up: rapid check questions

Pacing note — read this before you start: seven topics in 45 minutes is genuinely tight. The framework (Electronegativity & Bond Character, 3–10 min) is the one section not to cut — everything after it leans on it. If you're behind by the 24-minute mark, shorten the covalent-naming *teaching* (24–34) and lean on the interactive Module 4 quiz itself to do the teaching instead (it shows the rule with every answer) — that reclaims a few minutes. If still behind, alloys (inside the metallic section) is the safest full cut — it was already covered in depth in the previous session file, so it's genuine review, not new material.

Warm-up (3 min)

1. “What’s an ionic bond, in one sentence?” (*fast recall check from the test — should be automatic*)
2. **Hook:** “You know ionic bonds and you’re about to learn covalent bonds. Here’s the twist: they’re not two separate boxes — they’re two ends of the same spectrum. One number tells you which one you’re looking at. Let’s find out what that number is.”

The Framework — Electronegativity & Bond Character (7 min)

Recap definition (from the trends session): **electronegativity** is the ability of an atom in a compound to attract shared electrons to itself (Pauling scale, F = 4.0 highest).

The core idea (NCERT Class 11, section 4.3.6): no real bond is 100% ionic or 100% covalent — it’s a continuum, and the electronegativity *difference* (ΔEN) between the two bonded atoms tells you where a bond sits on it.

- **$\Delta\text{EN} = 0 \rightarrow$ nonpolar covalent bond.** When two *identical* atoms bond (H_2 , O_2 , Cl_2 , N_2 , F_2), the shared electron pair sits exactly between the two nuclei — neither atom pulls harder.
- **Small-to-moderate $\Delta\text{EN} \rightarrow$ polar covalent bond.** Textbook’s own example: H-F. Fluorine (EN 4.0) pulls the shared pair toward itself more than hydrogen (EN 2.1) does. The pair is still *shared*, just unevenly — this gives the molecule a **dipole moment** (a partial positive end, $\delta+$, and partial negative end, $\delta-$).
- **Large $\Delta\text{EN} \rightarrow$ ionic bond.** The pull is so lopsided the electron transfers completely rather than staying shared — this is exactly the Na/Cl picture from the ionic-bonding topic.

Working rule of thumb for placing a bond on the spectrum (a general-chemistry convention, not a number NCERT states — say this explicitly, since the textbook describes the change qualitatively, not with a cutoff): $\Delta\text{EN} < 0.5 \rightarrow$ nonpolar covalent · $0.5\text{--}1.7 \rightarrow$ polar covalent · $> 1.7 \rightarrow$ mostly ionic.

Worked example: H-F: $\Delta\text{EN} = |2.1 - 4.0| = 1.9 \rightarrow$ past the 1.7 line, yet H and F are both non-metals sharing electrons — so it’s *strongly polar covalent*, not truly ionic. This is exactly why the line is a guide, not a wall: bond character changes continuously, and real compounds don’t sort neatly into three boxes.

This is the lens for the rest of today — every bond you look at next, ask “what’s the ΔEN telling me here?”

Metallic Bonds, Properties of Pure Metals, Alloys (5 min)

Interesting wrinkle worth raising: ΔEN between two identical metal atoms is also 0 — same as H_2 . But two metal atoms don’t behave like H_2 . Why not? Because metallic bonding isn’t a *localized* shared pair between two atoms at all.

Metallic bond (standard chemistry model — not covered in either NCERT chapter, flagged here as before): metal atoms release their valence electrons into a shared, mobile “sea” of delocalized electrons; what remains is a lattice of positive metal ions held together by electrostatic attraction to that electron sea.

Properties of pure metals, explained by the model (NCERT Class 10 Ch3):

- **Malleable** (hammers into sheets) and **ductile** (draws into wires) — ion layers slide past each other while the electron sea flows around them; the bond doesn’t break.
- **Good conductors of heat and electricity** — the delocalized electrons carry charge and energy freely.
- **Lustrous** — free electrons reflect light.
- **Sonorous** — pure metals produce a ringing sound when struck (this is why school bells are cast from metal alloys, not non-metals).
- **Generally high melting points** — strong electrostatic attraction between ions and electron sea.

Alloys (brief — full depth was already covered last session): a homogeneous mixture of two or more metals, or a metal and a non-metal, made deliberately to improve properties (strength, hardness, corrosion resistance). Quick recall set: brass (Cu+Zn), bronze (Cu+Sn), stainless steel (Fe+Ni+Cr), solder (Pb+Sn). Alloys generally conduct less well and melt at a lower temperature than the pure metal.

Ionic Bonds, Ionic Ratios & Polyatomic Ions (9 min)

Recap via the framework: an ionic bond sits at the large- Δ EN end of the spectrum — electron transfer, not sharing. Na (EN 0.9) + Cl (EN 3.0): Δ EN = 2.1, well past the ionic line.

Ionic ratio of compounds — the criss-cross method (recap from the test):

1. Write the cation symbol first, then the anion symbol, with charges underneath.
2. Criss-cross the charge *numbers* into subscripts.
3. Simplify subscripts by their common factor, if any (e.g. $\text{Mg}^{2+} + \text{O}^{2-} \rightarrow \text{Mg}_2\text{O}_2 \rightarrow \text{MgO}$).

Polyatomic ions — ratios and writing: same criss-cross method, treating the whole polyatomic ion as one unit. Use **brackets** whenever 2+ of the *same* polyatomic ion appear:

Compound	Cation	Anion	Formula
Calcium carbonate	Ca^{2+}	CO_3^{2-}	CaCO_3 (2:2 simplifies to 1:1, no brackets needed)
Magnesium hydroxide	Mg^{2+}	OH^-	$\text{Mg}(\text{OH})_2$ (2 hydroxide ions — brackets required)
Aluminium sulfate	Al^{3+}	SO_4^{2-}	$\text{Al}_2(\text{SO}_4)_3$ (3 sulfate ions — brackets required)
Ammonium sulfate	NH_4^+	SO_4^{2-}	$(\text{NH}_4)_2\text{SO}_4$ (2 ammonium ions — brackets on the cation this time)

Quick recall table (already solid from the test — trigger, don't re-teach):

Polyatomic ions		Tricky monoatomic ions	
OH^- hydroxide, NO_3^- nitrate	CO_3^{2-} carbonate, SO_4^{2-} sulfate	Fe^{2+} ferrous / Fe^{3+} ferric	Cu^+ cuprous / Cu^{2+} cupric

Covalent Bonds + Naming with Prefixes (10 min)

Definition: a covalent bond forms when two atoms *share* one or more electron pairs, rather than transferring them — the low- Δ EN end of the spectrum. Already touched on via Ch9 9.4.1 (grade-9 foundation); today's focus is *naming* covalent compounds correctly.

The Greek prefix system (NCERT Class 9, Ch9, section 9.4.2):

1	2	3	4	5	6	7
mono-	di-	tri-	tetra-	penta-	hexa-	hepta-

The rules, with verified textbook examples:

1. **First element** is named plainly — *mono-* is dropped for the first element (CO_2 = carbon dioxide, not "monocarbon dioxide"). It only gets a prefix when there's more than one atom of it (N_2O_4 = **d**initrogen tetroxide).
2. **Second element** always takes a prefix (*mono-* included) plus the *-ide* ending (CO = carbon

monoxide).

3. **Dropped vowel:** the final *a* or *o* of a prefix is dropped before “oxide” — mon**o**xide, tet**ro**xide, pent**o**xide (never “monooxide,” “tetraoxide,” “pentaoxide”). N_2O_5 = dinitrogen **pent**oxide.
4. **Hydrogen-first exception:** when hydrogen is the first element, it never takes a prefix, whatever the count — H_2S = hydrogen sulfide, *not* “dihydrogen sulfide.”
5. **Common-name exception:** a few compounds are known only by their common names, not the systematic ones — H_2O is “water” (not “hydrogen monoxide”), NH_3 is “ammonia” (its systematic name would be nitrogen trihydride).

Worked set to run through together: CO → carbon monoxide · CS_2 → carbon disulfide · PCl_3 → phosphorus trichloride · SF_6 → sulfur hexafluoride · N_2O_4 → dinitrogen tetroxide.

Making the formula from the name (reverse direction): read the prefixes directly as subscripts. “Phosphorus trichloride” → P gets no subscript (mono- implied, dropped), Cl gets 3 → PCl_3 . “Sulfur hexafluoride” → SF_6 .

Interactive Activity — “Trends to Bonds,” Modules 3 & 4 (8 min)

Open <https://claude.ai/artifact/JLvqSjiFUK7GfAfcCQVhgxg> on the iPad. (*Private artifact — share from the page’s Share menu first if it doesn’t open on Dhairya’s device.*) Modules 1 and 2 (trends explorer, ionic bond builder) are there for follow-up self-study — today’s live time goes to the two new ones:

Module 3 — Bond Character (3-4 min): Start on the H-F quick-pick (already loaded as the default/textbook example) — read the spectrum placement and the δ^+/δ^- diagram together. Then have him pick Na + Cl and predict where the marker lands *before* tapping. Finish with H + H to see the $\Delta\text{EN} = 0$ nonpolar case for contrast.

Module 4 — Naming Covalent (4-5 min): Run the 10-question quiz. Let him answer solo, out loud, before revealing — the tool shows the specific rule behind each answer, which does double duty as the teaching if time is short (see pacing note above).

Wrap-up — Cold-Call Check Questions (3 min)

1. “ H_2 has $\Delta\text{EN} = 0$. Is its bond ionic, polar covalent, or nonpolar covalent?” (*Nonpolar covalent — identical atoms, pair sits exactly between them.*)
2. “Name PCl_3 .” (*Phosphorus trichloride — no prefix on P, tri- on Cl.*)
3. “Why does H_2S not get called ‘dihydrogen sulfide’?” (*Hydrogen never takes a prefix when it’s the first element, regardless of count.*)
4. “Write the formula for ammonium sulfate.” (*$(\text{NH}_4)_2\text{SO}_4$ — two ammonium cations balance one sulfate’s 2- charge, so ammonium needs the brackets this time.*)

Notes for next session

Covalent bonding today was naming-focused (prefixes, formula writing) — the natural next step is the *structural* side of covalent bonds: Lewis structures and VSEPR theory (NCERT Class 11 Ch4, the section right before 4.3.6), which explains *why* molecules have the shapes they do. This also sets up dipole moments properly (touched on today only for the H-F bond-character example).